Data-driven prediction of tribological performance (CoF and wear rate)

Trimestre Recherche SOPH.IA DATA Research Report

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Abstract

Friction and wear account for a significant fraction of global energy losses and material degradation. Diamond-like carbon (DLC) coatings are widely utilized to mitigate these effects due to their low friction and exceptional wear resistance. However, predicting their friction coefficient (CoF) and wear rate remains a complex challenge due to non-linear dependencies on material properties, surface states, and experimental conditions. In this study, a tribo-informatics approach is proposed to predict the dry CoF and wear rate of DLC coatings using heterogeneous literature data. A subset of variables was selected based on tribological relevance and data quality. An extensive preprocessing strategy was implemented, including categorical grouping, logarithmic scaling of highly dispersed variables, and the removal of physically inconsistent outliers. Among the machine learning pipelines evaluated, Extra Trees proved to be the most robust. Optimized models achieved $R^2 = 0.616$ for the CoF and $R^2 = 0.807$ for the wear rate. Feature importance analysis confirms that while operational parameters dominate the predictions, material descriptors provide essential complementary information. This work demonstrates the predictive potential of literature-derived datasets through targeted feature selection and tree-based ensemble methods.

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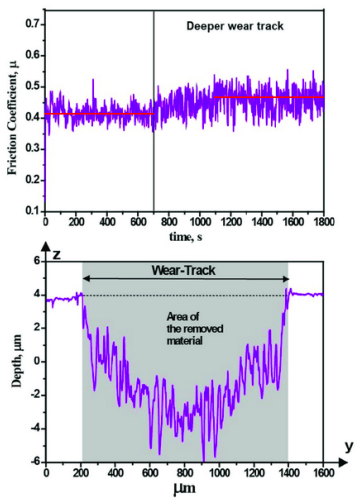
1 Introduction

Friction and wear critically affect performance, reliability, and energy efficiency of mechanical systems. Tribological losses are estimated to account for a significant fraction of global energy consumption, while wear-induced degradation drives maintenance costs and component failures in industries such as automotive engineering, aerospace, biomedical devices, and manufacturing [10]. Surface engineering strategies, particularly protective coatings, have emerged as essential tools for controlling tribological behavior.

Tribological performance is quantified using the friction coefficient (CoF) and the wear rate. The friction coefficient is defined as $\mu = \frac{F_\mathrm{t}}{F_\mathrm{n}}$ where F_t is the tangential friction force and F_n is the applied normal load [2]. Experimentally, it is computed and then calculeted as a mean value, as swhown in Figure 1.

Figure 1: CoF evolution & Wear Profile



The wear rate follows Archard's formulation: $k = \frac{V}{F_\mathrm{n}\, s}$ with V being the worn volume and s the sliding distance [1]. These metrics capture energy dissipation and material degradation.

Diamond-like carbon (DLC) coatings are widely employed in dry or boundary-lubricated contacts, with performance influenced by mechanical properties, surface morphology, chemical composition, and operating conditions such as load, temperature, and sliding velocity [4, 5]. Doping and hybrid structures are often used to further optimize friction and wear behavior [16].

Predicting CoF and wear rate remains challenging due to non-linear, multiscale interactions between surface asperities, tribochemical reactions, and environmental effects [11]. Literature-derived datasets provide broad coverage but introduce heterogeneity, missing values, and experimental noise. Addressing these challenges requires robust preprocessing, careful feature selection, and interpretability tools to ensure physically meaningful predictions [8].

In this study, a subset of input variables retained with the most interest was chosen based on tribological relevance and data completeness. This selection includes operational parameters, mechanical properties, chemical content, and structural descriptors, all considered to be highly influential on friction and wear behavior.

2 Methodology

The predictive modeling workflow is structured into three main stages: (i) data preparation and selection of the most relevant inputs, (ii) definition of the Extra Trees regression pipeline, and (iii) performance evaluation using standard metrics and interpretability tools.

2.1 Input Data Analysis and Feature Selection

Prior to model training, a detailed analysis of the input variables was conducted in order to characterize the nature of the dataset and guide feature selection. The compiled tribological database, built from heterogeneous literature sources, exhibits a significant proportion of missing values, reflecting differences in experimental setups, reporting standards, and measurement availability across studies.

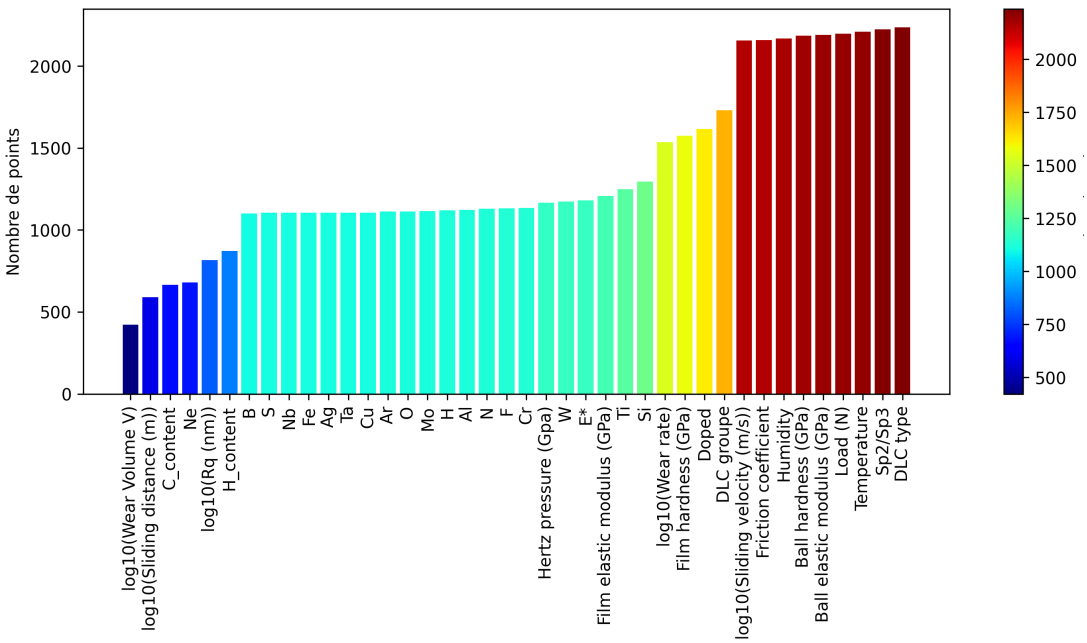


Figure 2: Number of available data points per selected input variable.

Rather than applying aggressive imputation strategies that could introduce artificial correlations, missing values were preserved whenever physically acceptable. This choice is justified by the use of tree-based models, such as Extra Trees, which are robust to incomplete data and can naturally handle mixed numerical and categorical inputs.

In addition, several continuous variables display strong right-skewed distributions spanning multiple orders of magnitude, a common characteristic of tribological data. To mitigate scale effects, reduce skewness, and improve model interpretability, logarithmic transformations (base 10) were applied to selected variables, notably wear rate, surface roughness, and sliding distance.

2.1.1 Selected Input Variables (Scenario 1)

Based on physical relevance, data availability, and redundancy analysis, a subset of input features was retained. These inputs capture the combined influence of operating conditions, mechanical properties, material chemistry, and surface characteristics on friction and wear behavior.

Operational Parameters

- `Sliding velocity` (m/s): Nominal relative speed between the contacting surfaces.
- `Temperature` (°C): Ambient or contact temperature during testing.
- `Humidity`: Environmental relative humidity, known to strongly affect DLC friction through adsorption layers and water-assisted lubrication mechanisms.
- `Sliding distance` (m): Total distance traveled during the test, reflecting cumulative damage and wear evolution. This variable is included as $\log_{10}(Slidingdistance)$ to account for its wide range of values.

Material and Chemical Descriptors

- `DLC group`: Categorical classification of the coating into *a-C*, *ta-C*, *a-C:H*, or *ta-C:H*. This grouping captures major structural differences in bonding, density, and hydrogenation, which are known to drive tribological behavior.
- `H content`: Hydrogen fraction within the DLC film, influencing hardness, elastic modulus, and friction through passivation and tribochemical effects.

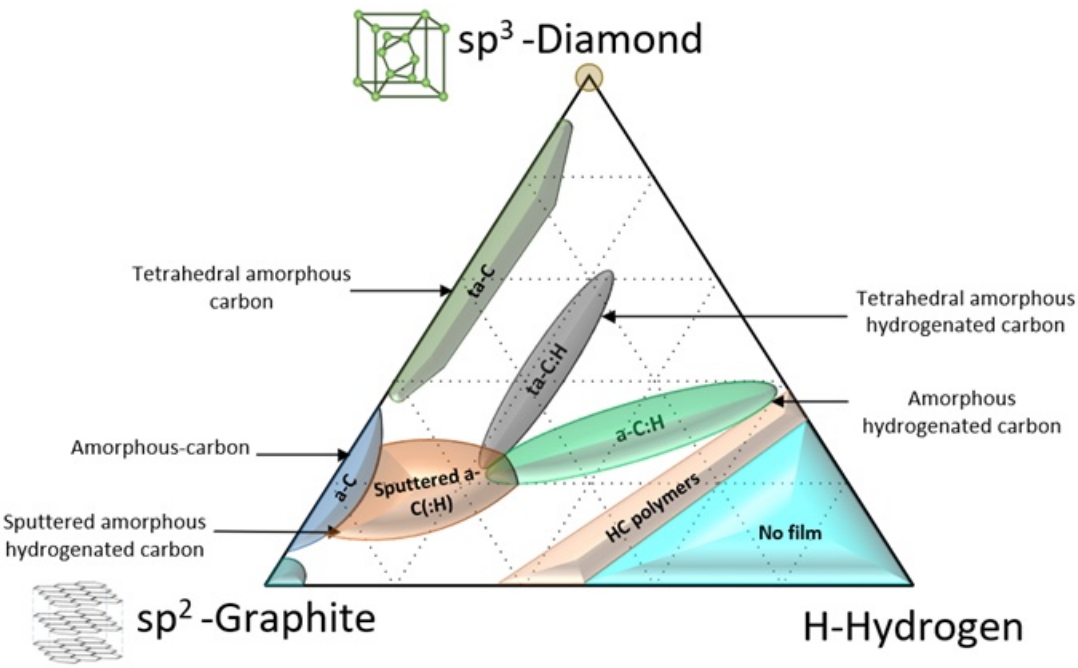


Figure 3: Ternary diagram representing the relative fractions of sp^2, sp^3 hybridization and hydrogen in DLC coatings.

- `Doped`: Element(s) with which the DLC is doped, such as Si, Ti, W, or Mo, which are often introduced to tailor mechanical or tribological performance.
- `Doped content`: Total dopant concentration.
- `sp3 %` and `sp2 %`: Fraction of carbon atoms in sp^3 and sp^2 hybridization states, respectively. These structural descriptors are directly linked to hardness, elasticity, and frictional response.

Mechanical Properties

- `Film hardness` (GPa): Resistance of the DLC coating to plastic deformation, strongly correlated with wear resistance.
- `Film elastic modulus` (GPa): Elastic stiffness of the coating.
- `Ball hardness` (GPa): Hardness of the counterbody, affecting wear mechanisms and transfer layer formation.
- `Ball elastic modulus` (GPa): Elastic modulus of the counterbody material (a round ball for every test)
- `Hertz contact pressure` (GPa): Maximum contact pressure calculated from load, geometry, and elastic properties, providing a physically meaningful measure of local contact stress.

Surface Roughness

- R_q (nm): Root-mean-square surface roughness of the coating. Due to its highly skewed distribution, this variable is included as $\log_{10}(R_q)$, emphasizing relative nanoscale differences that are critical for friction initiation.

2.1.2 Output Variables and Distribution Characteristics

Two primary output variables were considered in this study: the friction coefficient and the wear rate. Exploratory data analysis revealed distinct statistical behaviors for each.

The friction coefficient exhibits a moderately skewed distribution, closer to a logarithmic-type spread, reflecting the multiplicative influence of environmental and material factors. In contrast, the wear rate spans several orders of magnitude and displays a near-normal distribution when expressed in logarithmic scale. Consequently, the wear rate was systematically modeled as $\log_{10}$(wear rate), which stabilizes variance and aligns with the physical interpretation of wear as an order-of-magnitude phenomenon.

Overall, the combination of careful input selection, logarithmic transformations, and minimal preprocessing preserves the physical meaning of the variables while ensuring compatibility with data-driven modeling approaches.

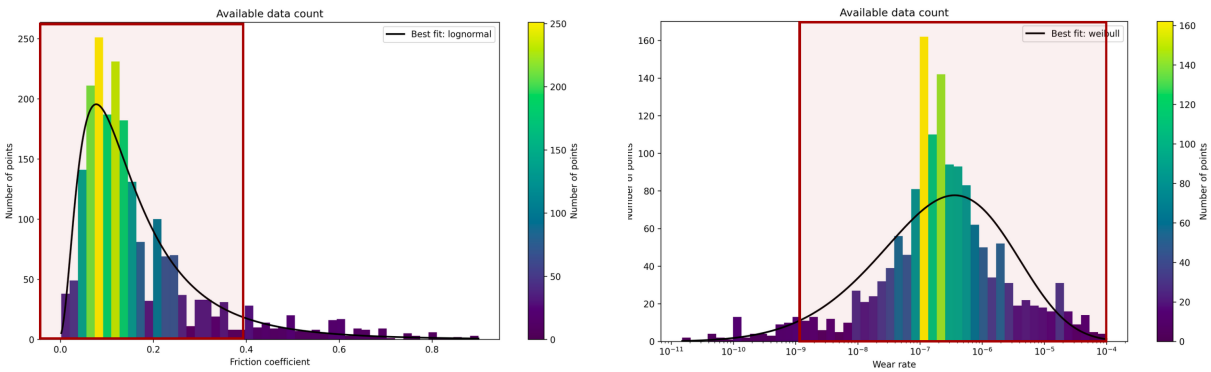


Figure 4: Distributions of (a) friction coefficient and (b) wear rate in the compiled literature dataset.

2.2 Extra Trees Regression Pipeline

The Extra Trees model was implemented using a pipeline combining categorical encoding and regression. Categorical inputs such as DLC group and doping presence were encoded with Target Encoding, optimized with a smoothing parameter.

Hyperparameter optimization was performed using Optuna, maximizing the R^2 score with 5-fold cross-validation. Key parameters explored include:

- Number of estimators: 300–1200
- Maximum depth: 5–40
- Minimum samples per split: 2–20
- Minimum samples per leaf: 1–10
- Maximum features per split: {sqrt, log2, None}
- Bootstrap: True or False

The pipeline was trained separately for each target: CoF and $\log_{10}$(Wear rate), ensuring optimized performance for both properties.

2.3 Evaluation Metrics

Model performance was quantified using three standard metrics:

2.4 Visual Evaluation

Prediction quality was assessed with scatter plots comparing predicted values (y_{pred}) against true values (y_{true}) for both CoF and wear rate. These plots provide a direct visualization of model accuracy and dispersion.

Permutation importance analysis was also performed to rank the contribution of each input variable to the model's predictive power. Operational parameters were found to dominate, with material descriptors providing complementary contributions.

Table 1: Regression metrics used to evaluate the Extra Trees model.

Name	Definition	Formula
Coefficient of determination (R^2)	Proportion of variance explained by the model	$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$
Root Mean Squared Error (RMSE)	Average magnitude of prediction errors	$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$
Mean Absolute Error (MAE)	Average absolute difference between predictions and true values	$MAE = \frac{1}{n}\sum_{i=1}^{n}\lvert y_i - \hat{y}_i \rvert$

3 Results

This section presents the predictive performance of the Extra Trees model. The results highlight the model's ability to generalize from heterogeneous data after a rigorous **Bayesian hyperparameter optimization** process.

3.1 Hyperparameter Optimization and Convergence

The optimization was conducted using **Optuna**, aiming to maximize the R^2 score over 100 trials. Figures 5 illustrates the evolution of the R^2 during the process. For both targets, we observe that the performance **reaches a plateau** relatively quickly. This saturation suggests that the model has effectively captured the available information density within the current feature set, and further tuning of hyperparameters yields diminishing returns.

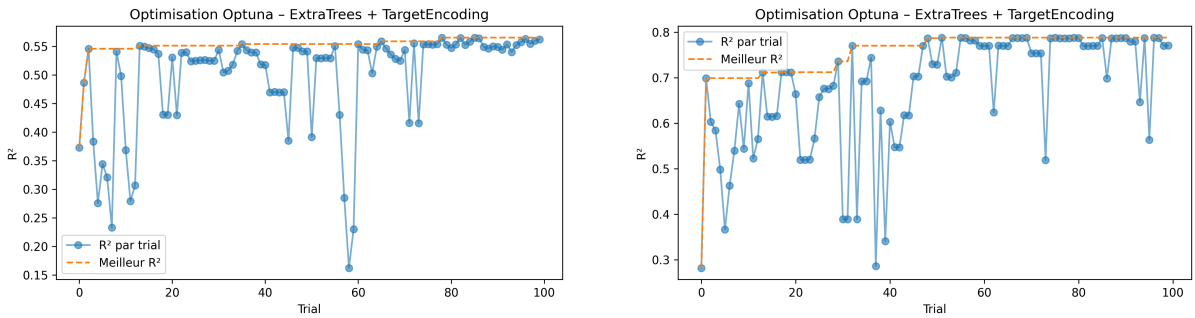


Figure 5: Optimization history showing the R^2 evolution for (a) CoF and (b) Wear rate. The convergence toward a plateau indicates model saturation.

The best parameters identified for each target are summarized in Table 2. Notably, the **smoothing parameter** for Target Encoding differs significantly between the two models, reflecting the different scales and distributions of the target variables.

Table 2: Best hyperparameters identified via Optuna optimization.

Hyperparameter	Friction Coefficient	log10(Wear Rate)
n_estimators	926	835
max_depth	30	26
min_samples_split	3	3
min_samples_leaf	1	1
max_features	None	None
bootstrap	True	False
smoothing	2.576	41.951

3.2 Final Prediction Performance

The evaluation was performed on a **hold-out test set** representing 20% of the total data. The final metrics demonstrate a strong correlation for the wear rate and a satisfactory fit for the CoF.

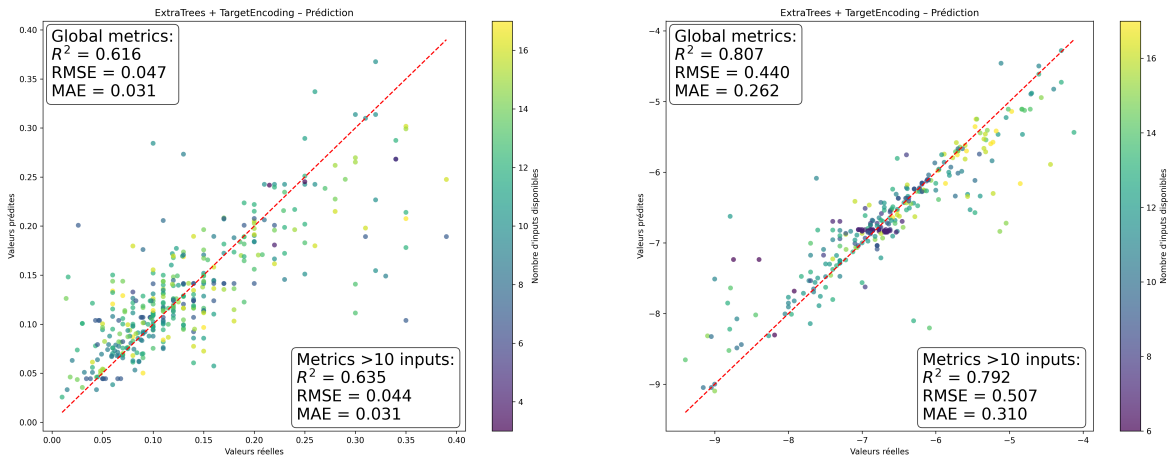


Figure 6: Predicted vs. true values for CoF (left) and wear rate (right) using the Extra Trees model. Metrics: R^2, RMSE, and MAE.

3.2.1 Coefficient of Friction (CoF)

The model captures the **global trends** effectively, although the moderate R^2 of 0.616 reflects the inherent noise and experimental variability found in multi-source literature data.

Table 3: prediction metrics for the friction coefficient.

Metric	R^2	RMSE	MAE
Value	**0.616**	0.047	0.031

3.2.2 Wear Rate

The performance for the wear rate is notably higher ($R^2 = 0.807$), indicating that the **logarithmic transformation** successfully linearized the primary relationships between inputs and material loss.

Table 4: prediction metrics for wear rate.

Metric	R^2	RMSE	MAE
Value	**0.807**	0.440	0.262

3.3 Permutation Feature Importance

To assess the effect of individual inputs on model predictions, permutation importance was computed on the test set to evaluate the contribution of each input variable. Figure 7 show variables affecting CoF and wear rate predictions the most.

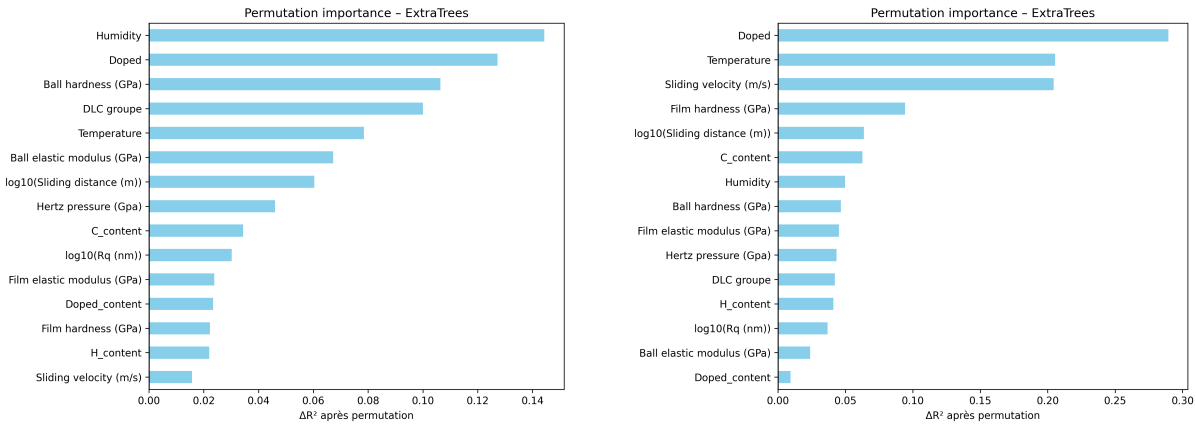


Figure 7: Permutation importance for CoF (left) and wear rate (right) predictions.

The top 3 most important variables for CoF are **Humidity**, **Doped**, **Ball hardness (GPa)**. While top 3 for Wear rate is **Sliding velocity (m/s)**, **Doped**, **Temparature**. While these trends are physically plausible and consistent with tribological expectations, a detailed interpretation of the underlying mechanisms would require dedicated experimental or physics-based analyses and is therefore beyond the scope of this study.

3.4 Summary

The Extra Trees model, trained on the most relevant input features, achieved relatively interesting predictive performance:

- Coefficient of friction: $R^2 = 0.616$, RMSE $= 0.047$, MAE $= 0.031$.
- Wear rate: $R^2 = 0.807$, RMSE $= 0.428$, MAE $= 0.260$.

The scatter plots of predicted versus true values confirm that the model captures the general trend in the dataset, while permutation importance and sensitivity analysis highlight the dominant role of operational parameters and secondary influence of material descriptors. Overall, Extra Trees provides a robust, interpretable, and physically consistent framework for predicting tribological performance from literature-derived data.

4 Conclusions and Perspectives

This study confirms the viability of using **Extra Trees** to navigate the complexities of literature-derived tribological data. By focusing on a refined subset of physical descriptors, we achieved predictive models for DLC coatings with R^2 scores of **0.616** (CoF) and **0.807** (Wear rate).

While these results are promising, they also highlight the limitations of current data-driven approaches in tribology. The "plateauing" observed during optimization suggests that we are approaching the **information limit** of the current feature set. The moderate accuracy for CoF, in particular, reminds us that friction is a highly sensitive system response rather than a static material property.

Perspectives for improvement

To surpass these performance limits, future work will focus on:

- **Feature Engineering:** Creating new, more **physically coherent inputs**, such as combined dimensionless numbers to better represent the contact mechanics.
- **Data Enrichment:** Expanding the dataset to include more specific details on **surface chemistry** and environment, which are often overlooked in broad literature surveys.
- **Hybrid Modeling:** Integrating simplified **analytical laws** into the ML pipeline to guide the model in regions where data is sparse.

In conclusion, this work provides a robust baseline for **data-driven coating design**, offering a path toward more efficient materials selection and operational optimization in mechanical engineering.

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A Appendix: Additional Methods and Scenarios Tested

During the preliminary stages of this study, multiple data processing strategies, machine learning models, and feature scenarios were evaluated to assess their impact on prediction performance. These include:

- **Data Imputation and Scaling Methods:** Mean imputation, median imputation, K-nearest neighbors (KNN) imputation, and optional feature scaling were explored, although tree-based models showed best performance without imputation.
- **Regression Models:** In addition to Extra Trees, Random Forest, XGBoost, and Elastic Net models were tested. Extra Trees consistently outperformed the alternatives in predictive accuracy and robustness.
- **Feature Scenarios:** Multiple feature sets were initially considered, ranging from operational and mechanical parameters only, to extended sets including doping information and elemental compositions. The selected input set balances relevance, completeness, and interpretability.
- **Hyperparameter Optimization:** Optuna-based Bayesian optimization was applied to tune model hyperparameters, maximizing cross-validated R^2 scores.

These exploratory tests provided a systematic assessment of modeling options and informed the final choice of inputs, pipeline, and model presented in the main report. Although detailed results for all alternative methods and scenarios are not shown, they confirm that the selected approach—Extra Trees with the most relevant input variables—offers the best compromise between accuracy, robustness, and physical interpretability.